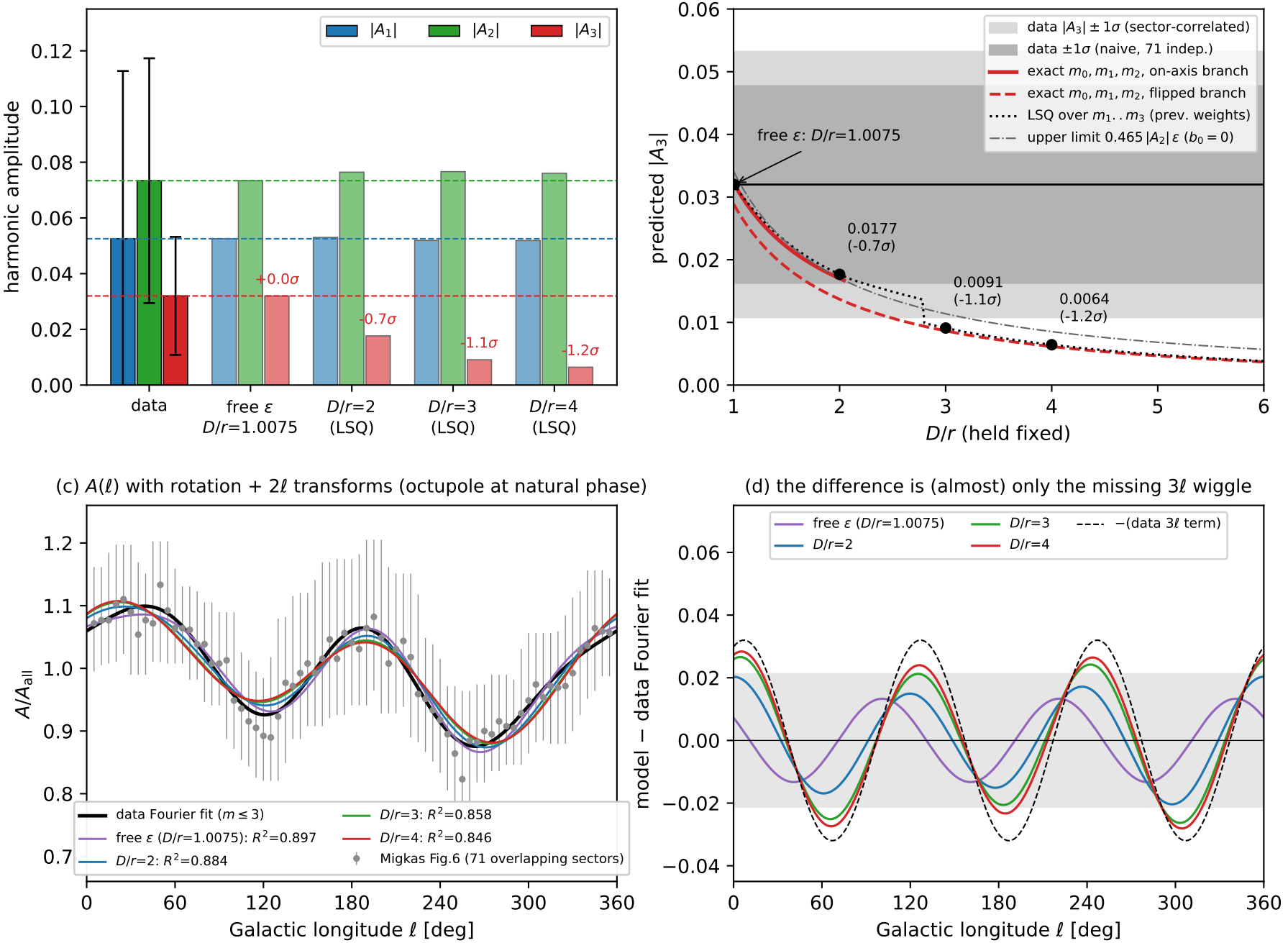
(a) amplitudes: model (pale) vs data (solid, ±1σ sector-correlated)
harmonic amplitude
|A1| |A2| |A3|
+0.0σ
-0.7σ
-1.1σ
-1.2σ
data
free ε D/r=1.0075
D/r=2 (LSQ)
D/r=3 (LSQ)
D/r=4 (LSQ)
(b) octupole predicted once m0, m1, m2 are matched
predicted |A3|
D/r (held fixed)
data |A3| ± 1σ (sector-correlated)
data ±1σ (naive, 71 indep.)
exact m0, m1, m2, on-axis branch
exact m0, m1, m2, flipped branch
LSQ over m1..m3 (prev. weights)
upper limit 0.465 |A2| ε (b0 = 0)
free ε: D/r=1.0075
0.0177 (-0.7σ)
0.0091 (-1.1σ)
0.0064 (-1.2σ)
(c) A(ℓ) with rotation + 2ℓ transforms (octupole at natural phase)
A/Aall
Galactic longitude ℓ [deg]
data Fourier fit (m ≤ 3)
free ε (D/r=1.0075): R²=0.897
D/r=2: R²=0.884
D/r=3: R²=0.858
D/r=4: R²=0.846
Migkas Fig.6 (71 overlapping sectors)
(d) the difference is (almost) only the missing 3ℓ wiggle
model − data Fourier fit
Galactic longitude ℓ [deg]
free ε (D/r=1.0075)
D/r=2
D/r=3
D/r=4
−(data 3ℓ term)